Encirra

A Coupled Power–Water Digital Twin for Verified Recovery from Cascading Urban Infrastructure Failures

See more. Recover faster. Safer cities for tomorrow.

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NEECON 2026 Undergraduate Innovation Challenge
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17:20 · Feeder F3 fault, Encirra recovery applied. Red: isolated faulted section. Green: cables re-energised by the verified plan. Amber: healthy supply. Cyan: water mains in the exposed underground. Labels and colours are driven by the power-flow and hydraulic solutions.

1 The problem: failures cross domains

Urban water networks depend on electrically driven pumps. When a distribution feeder trips, the **power** outage is visible immediately, but the **water** consequence is hidden: elevated tanks keep pressure up for tens of minutes, then critical facilities such as hospitals lose supply.

Conventional fault restoration maximises restored electrical load. It treats a pump as an ordinary load and counts a hospital on its standby generator as "served", so it can choose a plan that is electrically optimal and still lets the hospital run out of water.

72 min
from pump trip to hospital pressure < 20 m (simulated)

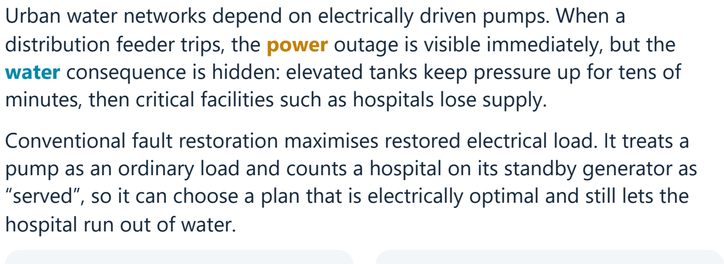
0 min
below service pressure with Encirra's verified plan

2 What Encirra is

A physics-driven, optimization-assisted digital twin that **models** the coupled networks, **predicts** the cascade, **evaluates** recovery plans across both domains and **independently verifies** them before they are recommended.

- **Electrical:** IEEE 33-bus distribution feeder (Baran & Wu) solved with AC power flow in pandapower; 37 switchable lines, 5 normally-open ties.
- **Hydraulic:** two-zone network (2 pumping stations, 2 elevated tanks, desalination supply) solved as a pressure-driven extended-period simulation with EPANET 2.2 via WNTR.
- **3D twin:** a Dubai-inspired district with an exposed underground; every building is mapped to the bus that supplies it.

3 Coupling the two models



$$P_{pump} = \rho \cdot g \cdot Q \cdot H / \eta$$

electrical demand from the hydraulic solution (W → kW), fed back into the AC power flow

If a pump's bus is de-energised or below 0.90 pu, the pump stops and the hydraulic model is re-solved, so flow, pressure and tank levels always come from the recalculated simulation, never from a scripted curve.

4 Recovery planning

After protection isolates the fault, Encirra generates radial reconfigurations from remote-operable switches (closing up to two ties, opening a sectionaliser, optionally shedding non-critical load) and simulates every candidate on the coupled model over the remaining horizon.

Strategy	Decision rule
None	Baseline: the outage persists.
Grid-only	Conventional restoration: maximise restored electrical energy within hard limits.
Encirra	Weighted objective over served load, critical load, water delivered, critical-facility water service, pressure/storage margin and switching cost.

5 Independent verification

The selected plan is re-simulated minute by minute on networks rebuilt through a separate code path. Only plans that pass every hard check are labelled **Simulation Verified**.

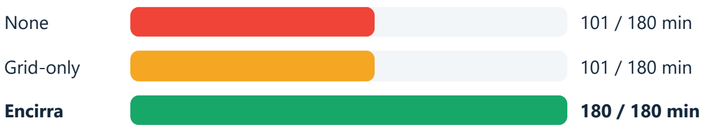
- ✓ Power-flow convergence
- ✓ Thermal line ratings
- ✓ Fault stays isolated
- ✓ Critical pressure ≥ 20 m
- ✓ Remote switching only
- ✓ Voltage band 0.90–1.05 pu
- ✓ Radiality & connectivity
- ✓ Pump supply available
- ✓ Tank above minimum
- ✓ Critical facility service

Feeder-fault scenario: **80 candidates** evaluated, **59 rejected** with their physical reason, e.g. **V(B29) = 0.775 pu < 0.90 pu** or **line B25–B29 loaded 130 % > 100 %**.

6 Results

Scenario	Strategy	Net restored	PS-2	Hospital min. p	Min < 20 m
Feeder fault	None	0 %	off	0.0 m	79
	Grid-only	69.2 %	off	0.0 m	79
	Encirra	**78.7 %**	**on**	**37.8 m**	**0**
Pump power loss	Grid-only	91.7 %	off	0.0 m	79
	Encirra	91.2 %	**on**	**37.8 m**	**0**
High demand	Both (same plan)	78.6 %	on	37.1 m	0

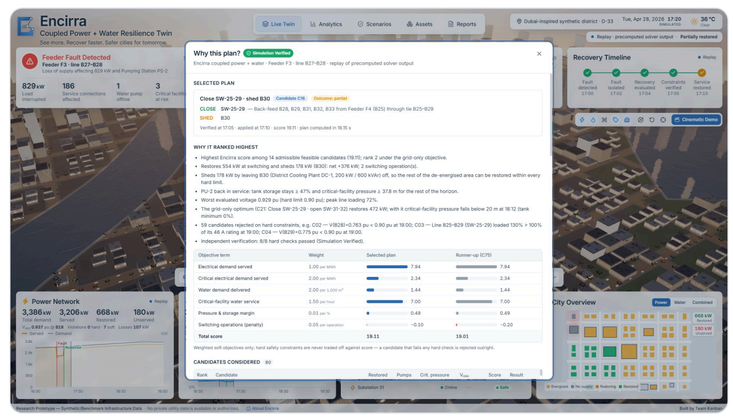
Feeder fault: hospital water service over the 3-hour horizon



Every selected plan passed every hard check. Under high demand both planners converge on the same plan, and Encirra's advantage depends on the synthetic tie-line capacity; both findings are reported with a sensitivity analysis.

7 The cascade, side by side

Grid-only at 18:20. Electrically verified, yet PS-2 stays dark: Tank T2 has emptied and hospital pressure is 0 m, 72 minutes after the trip.



"Why this plan?" Every candidate, its score breakdown, rejected alternatives with their failed checks, and the verification result.

8 Contribution & next steps

- An open, reproducible coupled power–water co-simulation with explicit pump-power feedback.
- Cross-domain recovery that protects critical water service, with independent verification and transparent evidence for every decision.
- Validated on a published benchmark (Vmin 0.913 pu at B18, 202.7 kW losses reproduced); 90 simulation, 45 interface and 15 browser tests.
- **Next:** valve actions in the planner, larger and real-topology networks, uncertainty in demand and repair times, and utility partnership for validation.

Built by Team Kanban

1. M. E. Baran, F. F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," *IEEE Trans. Power Delivery*, 1989.
2. L. Thurner et al., "pandapower — an open-source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Trans. Power Systems*, 2018.
3. K. A. Klise et al., "A software framework for assessing the resilience of drinking water systems to disasters with an example earthquake case study," *Environ. Model. Softw.*, 2017 (WNTR).
4. L. Rossman et al., *EPANET 2.2 User Manual*, U.S. EPA, 2020.

Live demo
encirra.kanbanstudios.ae

Source code
github.com/4waiz/Encirra